

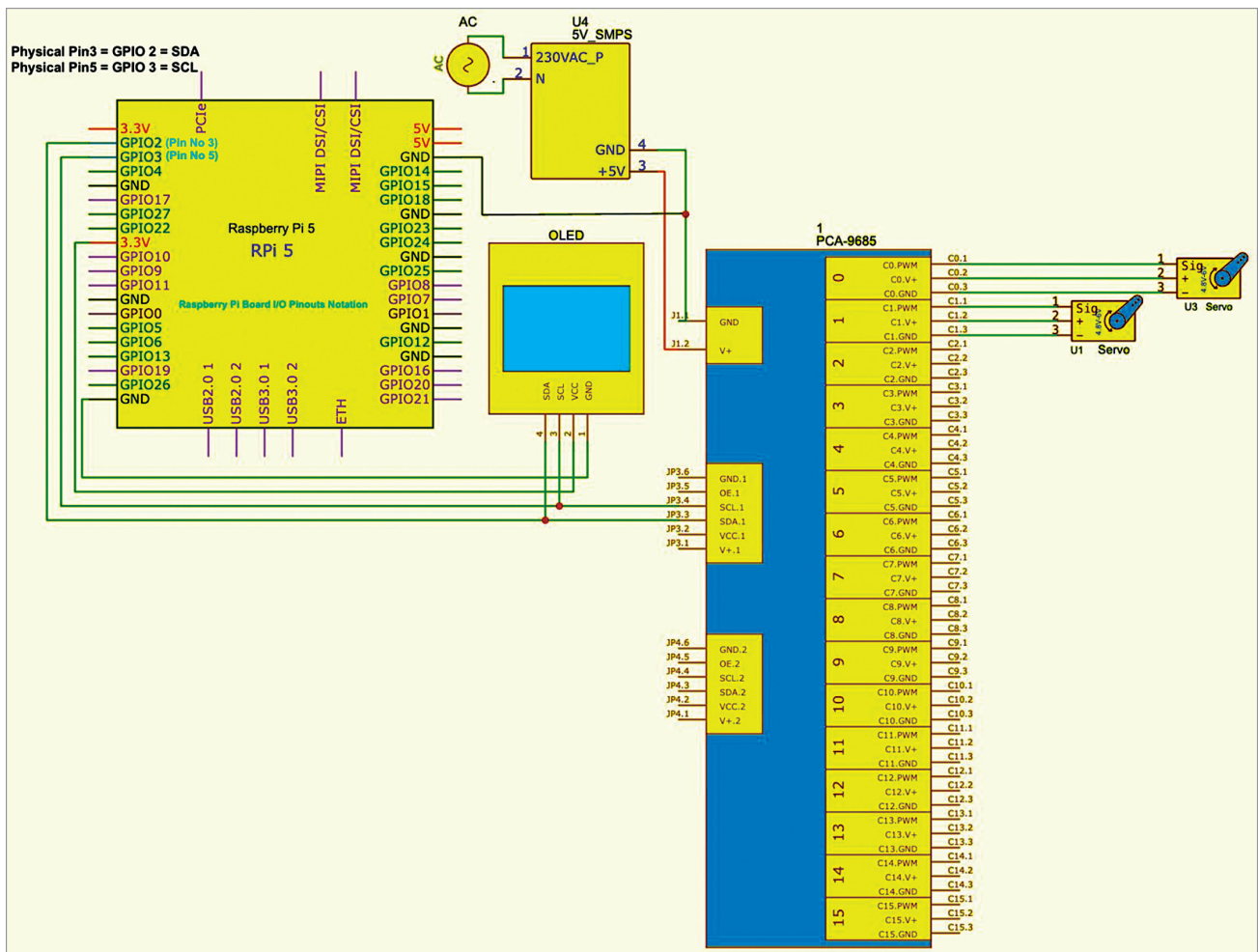


Fig. 6: Circuit diagram

```
$> python pi5_facetracking.py -help.
```

A typical command-line invocation:

```
$> python pi5_facetracking.py --labels-  
json personface_labels.json --input rpi  
cam --show-fps -headless
```

The `--input` option can accept different video sources:

`rplicam`→Raspberry Pi camera module

`/dev/video0`→USB webcam

`test_video.mp4`→Local video file

`test_video.webm`→A downloaded YouTube video

Additional options

`Show-fps`→Displays the frame-per-second (FPS) processing speed.

`Headless`→Runs the program without requiring a display connected to the Raspberry Pi 5.

When headless mode is enabled, video output is not displayed even if a monitor is connected. Only

text-based output, such as logs and print statements, appears on the console.

Raspberry Pi-5 code: Without Hailo

For operation without the Hailo-8L accelerator, the YOLO model `yolov8n-face-lindevs.pt` is used. This model performs reliably for face detection and delivers improved results under slightly lower-light conditions on Raspberry Pi 4 and Raspberry Pi 5 compared with several other YOLO models. The code runs at a moderate speed on both platforms with 8GB RAM.

The main file used in this configuration is `pi4_facetracking.py`. Before execution, activate the Python virtual environment and install the required modules:

```
$ source setup_env.sh # Installs and  
activates the Python virtual environment  
$ pip install ultralytics # Installs  
YOLO for Python  
$ pip install pillow # Installs  
Pillow for image handling
```

Important notes.

Ensure that I²C is enabled on the Raspberry Pi, as the system uses I²C-connected devices.

The pi4 filename prefix is

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